

Loyola Marymount University

# CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

# Announcements

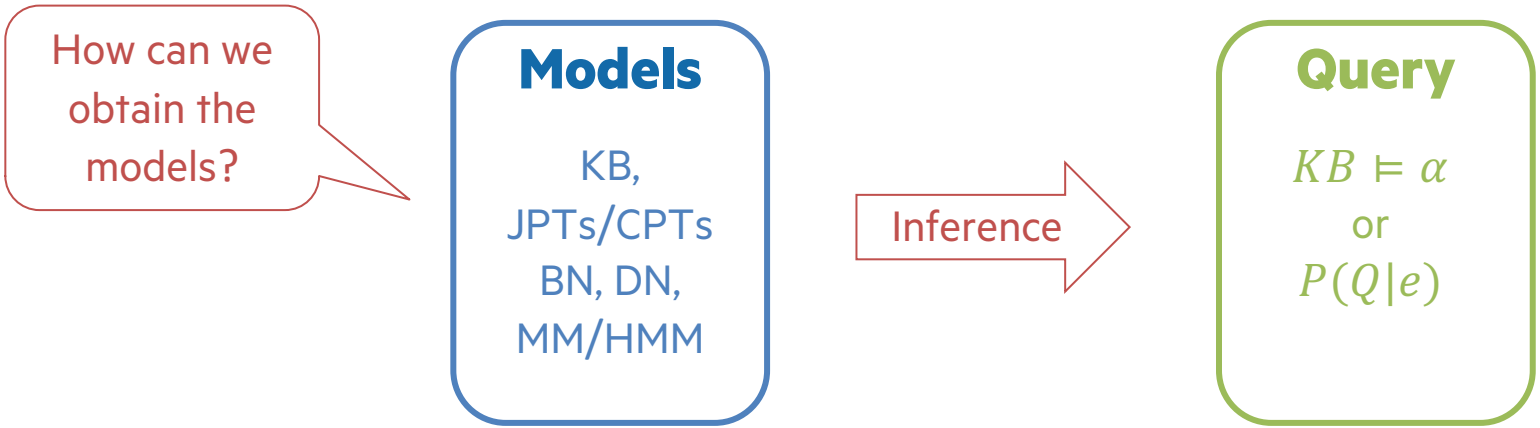
- Classwork 5
  - Available on Brightspace
  - Due on Friday, 11/8 @5pm
- Homework 3
  - Available on Brightspace after today's lecture
  - Due on Friday, 11/15 @5pm

# This Lecture

- Learning
  - What is learning
  - Types of learning
  - Learning from Examples/Data

# Inference

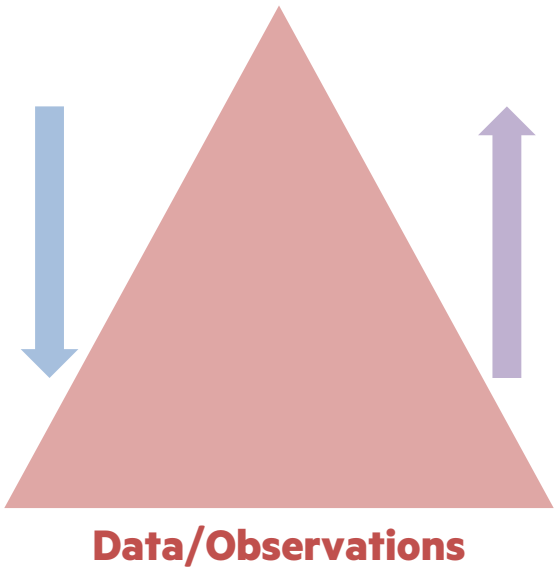
- What is inference?
  - Propositional: KB(rules+facts) -> new facts
  - Probabilistic: $P(Q|e)$



# Top-Down vs. Bottom-Up

## Top-Down

- **Model -> Data**
- Start from
  - programmer-curated **model**
  - general, high-level **concepts/variables**
- **Interpret** witnessed data through the model




## Bottom-Up

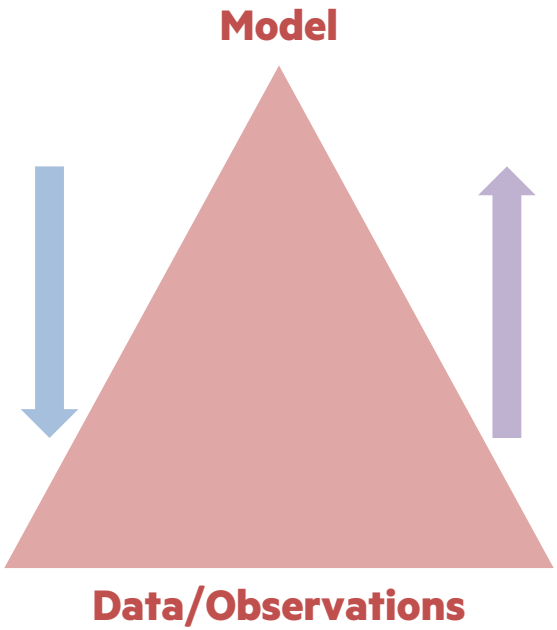
- **Data -> Model**
- Start with **data**
- **Learn** the model (i.e., structure relationships between variables)

What are the pros and cons of these approaches?

# Top-Down vs. Bottom-Up

## Top-Down

- **Pros**
  - Clear model structure
  - Intuitive for experts
  - Knowledge representation
- **Cons**
  - May be **biased** by preconceptions
  - May **not fit** all available data
  - Risk of **oversimplification**




## Bottom-Up

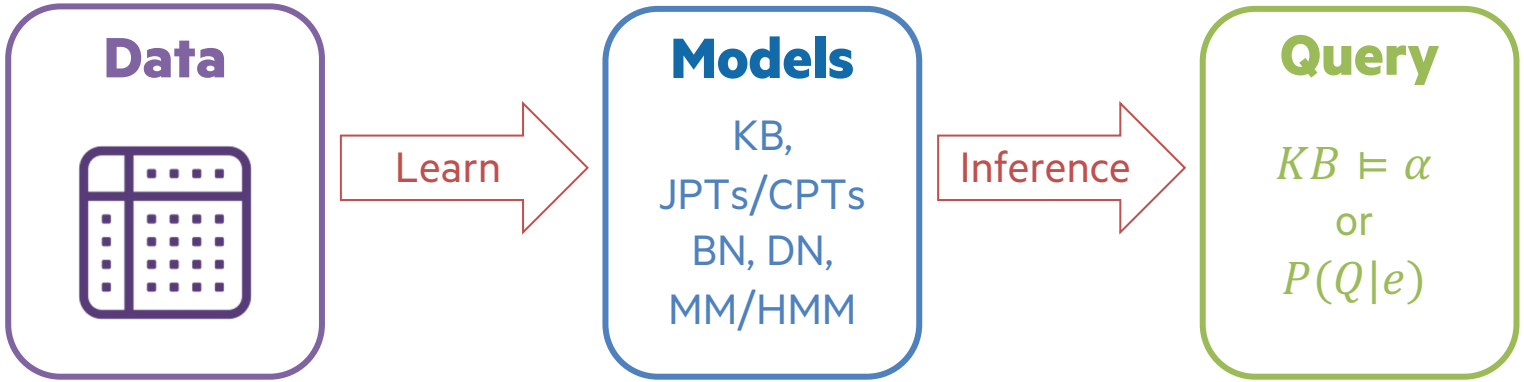
- **Pros**
  - Discover unexpected relationships
  - Pattern discovery
  - More objective
- **Cons**
  - Computationally intensive
  - Data intensive
  - Might find spurious correlations

# Learning

- What is learning?
  - **Learning** is the process of improving the agent's performance after making observations about the environment
  - Helps agents adapt to changing environments/new information
  - Synthesize memories
- **Machine learning** is the process of computationally discovering/constructing a model **from data / experience**.

# Learning and Inference

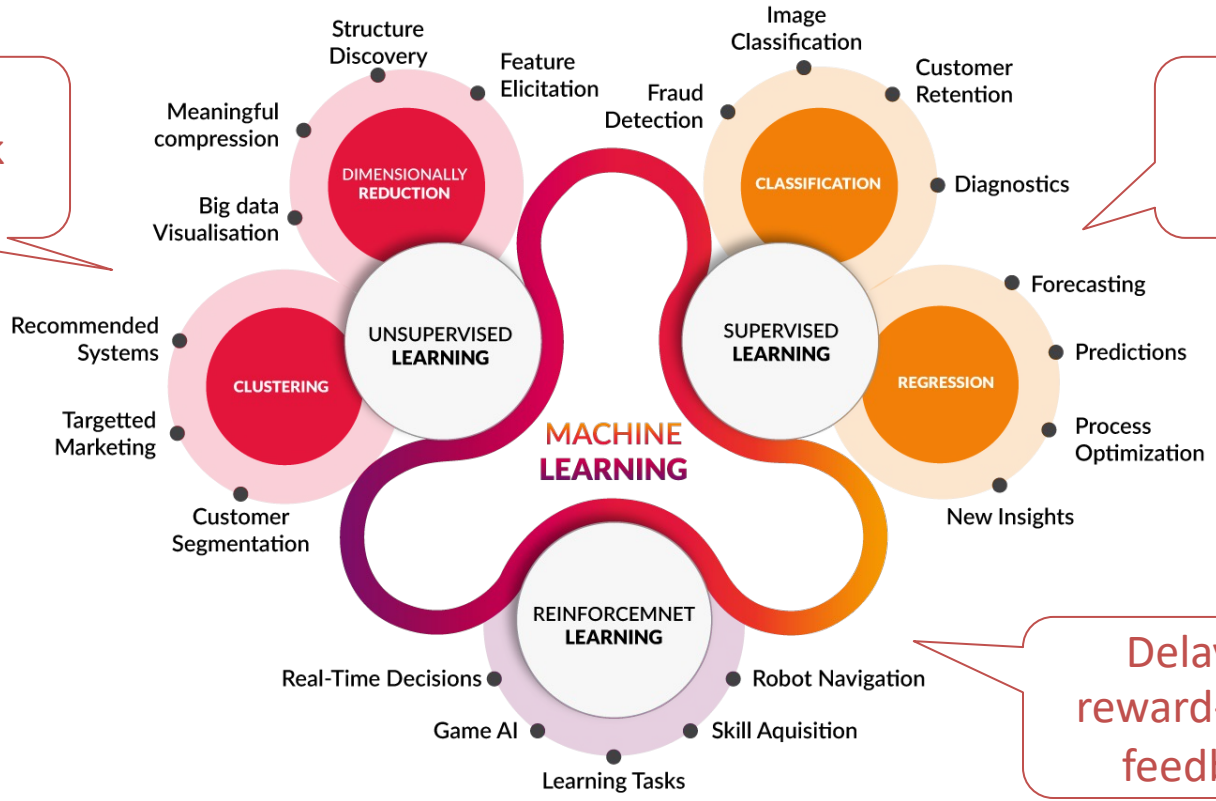
- Learning: “Model Development”
- Inference: “Model Application”



# Types of Learning – In Terms of Feedback

No feedback

Immediate & explicit feedback

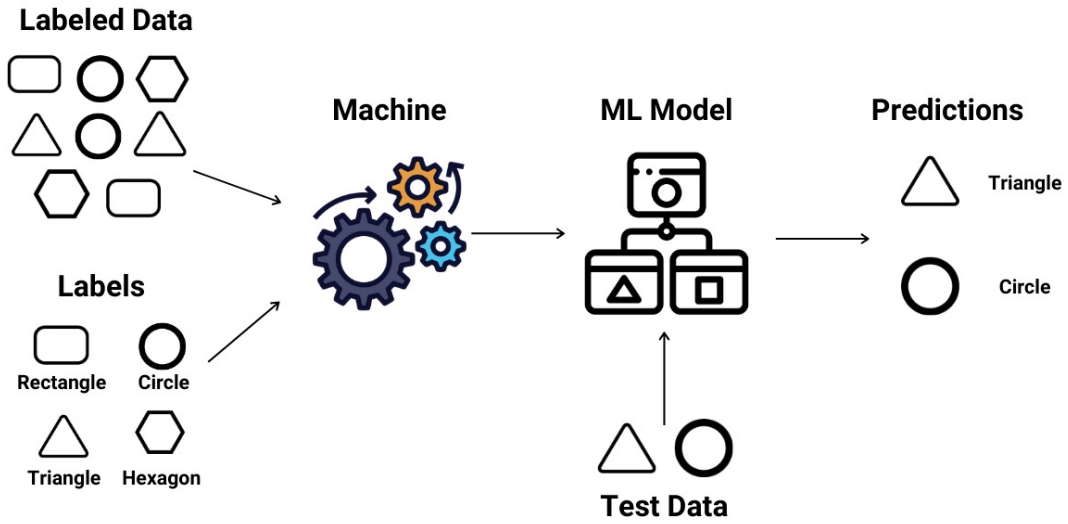



Delayed, reward-based feedback

Image Source: http://www.cognub.com/index.php/cognitive-platform/

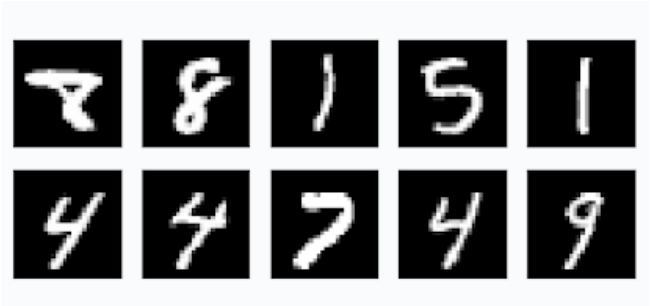
# Supervised Learning

- Data: Labeled data (input-output pairs)
- Goal: Find a mapping between inputs and outputs



# Supervised Learning

- Examples





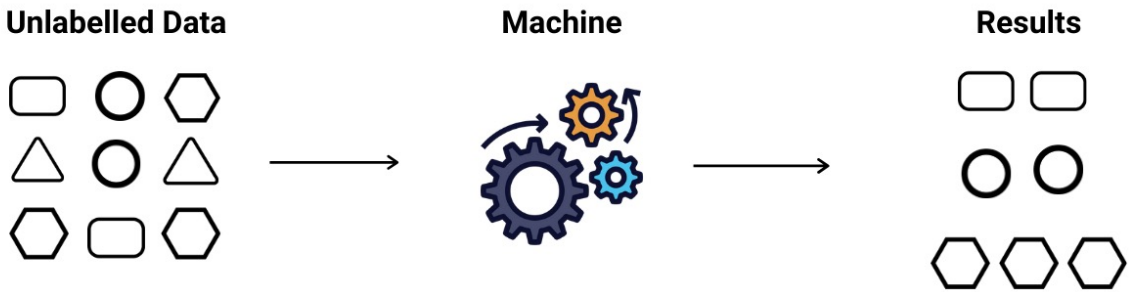






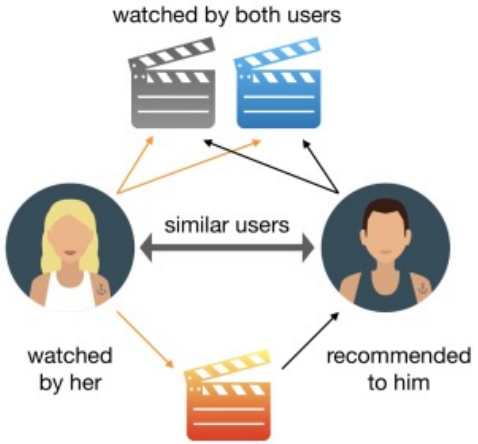

# Unsupervised Learning

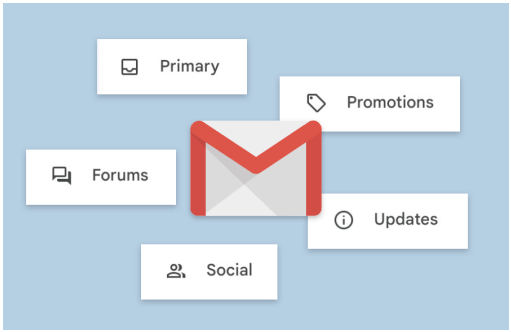
- Data: unlabeled data (no correct answers/targets)
- Goal
  - Find structure/patterns in data
  - Group similar items, reduce dimensionality



# Unsupervised Learning

- Examples

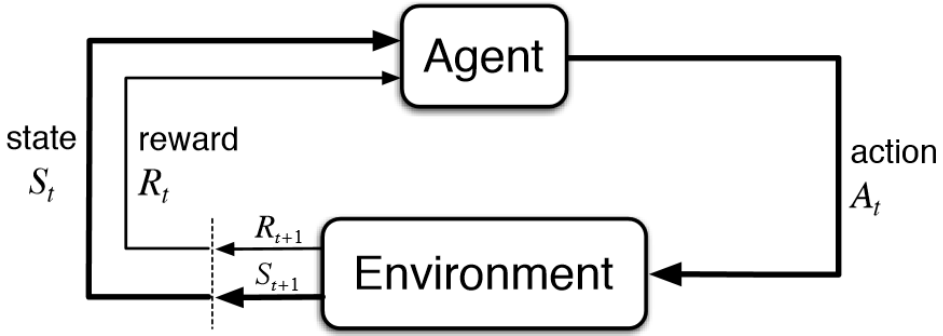


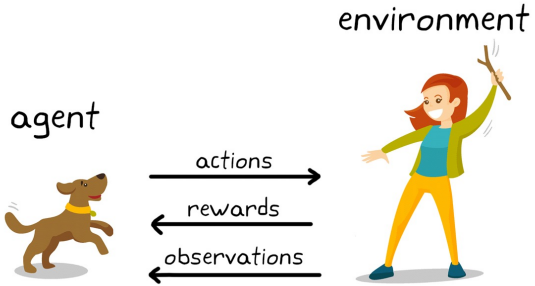






# Reinforcement Learning

- Data
  - No predefined data
  - Delayed feedback (rewards)
- Goal: maximize rewards by interacting with environment

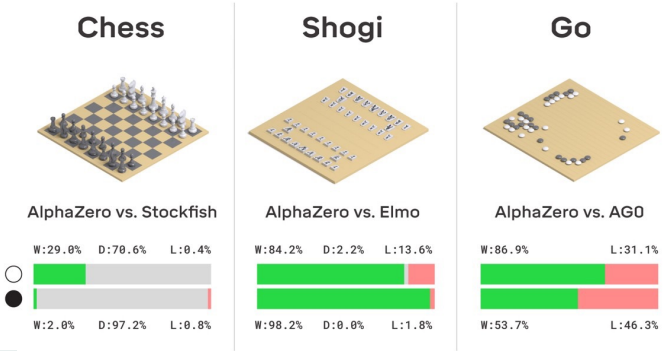


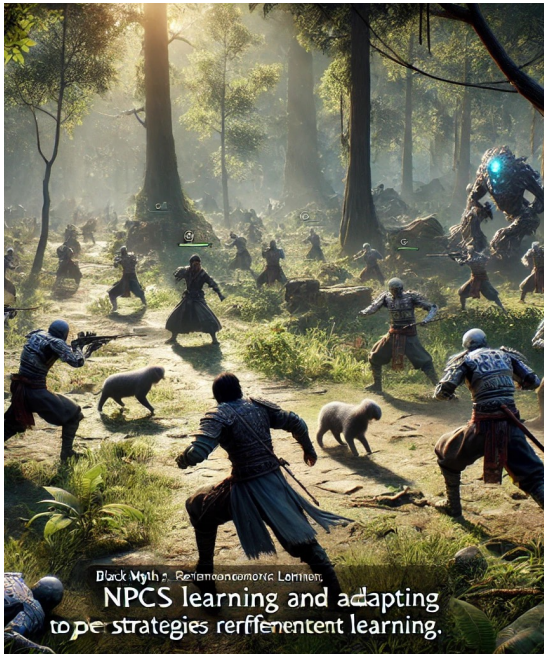



# Reinforcement Learning

- Examples










# Types of Learning

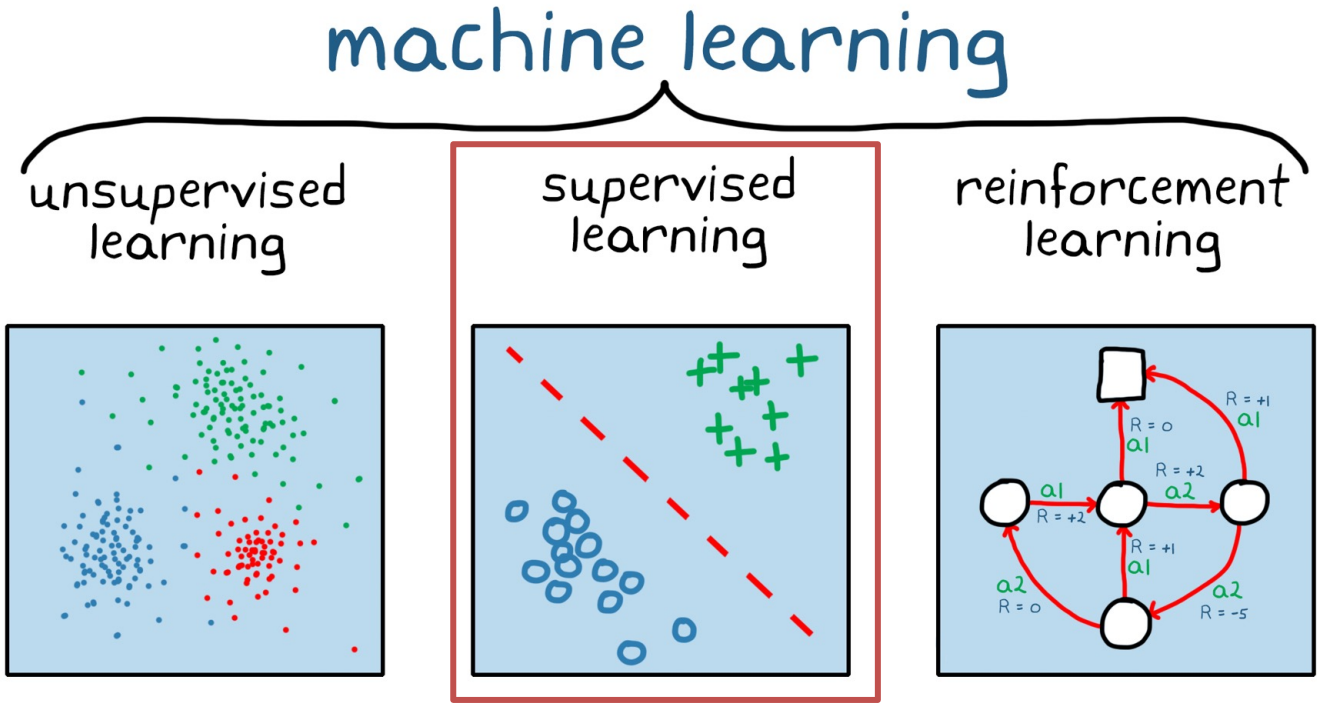
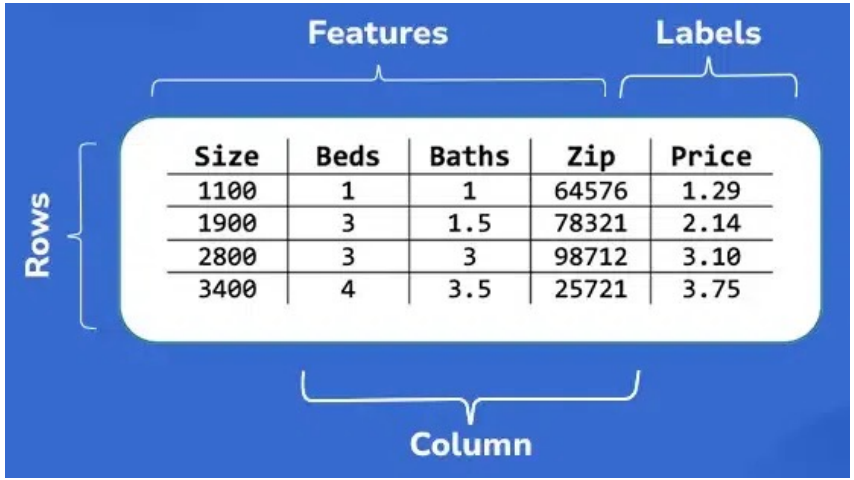



Image Source: https://au.mathworks.com/discovery/reinforcement-learning/

# Learning from Examples

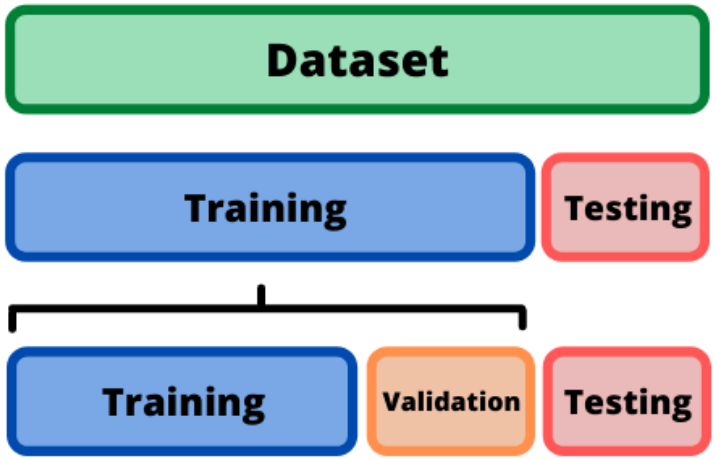
- What to “learn” from examples?
  - Functional mapping from input $X$ to output $Y$
- Supervised Learning
  - Features/Input $X$
  - Target/Output/Label $Y$




| Size | Beds | Baths | Zip | Price |
|---|---|---|---|---|
| 1100 | 1 | 1 | 64576 | 1.29 |
| 1900 | 3 | 1.5 | 78321 | 2.14 |
| 2800 | 3 | 3 | 98712 | 3.10 |
| 3400 | 4 | 3.5 | 25721 | 3.75 |

# Training, Validation, and Test

- Training (~70%)
  - Train/learn the model
- Validation (~20%)
  - Tune hyperparameters
  - Evaluate during training
- Test (~10%)
  - For final model evaluation
  - Unbiased estimate of model performance



# Overview of the Remaining Topics

- Machine Learning
  - Naïve Bayes Classifiers
  - Logistic Regression
  - Artificial Neural Networks
  - Intro to Deep Learning
  - Intro to Generative Models